

Cleanroom & HVAC Qualification Report

Illustrative facility qualification structure covering HEPA integrity, airflow, pressure, classification and recovery.

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1. Purpose and scope

This sample qualification report demonstrates a concise approach for a GMP cleanroom/HVAC system. It combines facility checks, HEPA integrity, airflow, room pressure, particle classification and recovery evidence into one controlled qualification conclusion.

2. Area and system boundary

- **Example area:** one Grade C preparation room and adjacent Grade D support room served by a dedicated air-handling unit.
- **System boundary:** AHU, terminal HEPA filters, supply/return paths, room pressure monitoring and environmental controls.
- **Qualification conditions:** at-rest classification unless otherwise defined, calibrated instruments, approved occupancy state and established cleaning status.

3. Qualification matrix

Test area	Evidence	Example acceptance
Installation / documentation	AHU ID, filter IDs, drawings, pressure instruments, calibration, maintenance status	Complete and traceable
HEPA integrity	Aerosol challenge and downstream scan of terminal filters	No local leak above approved criterion
Air volume / ACH	Supply airflow measurements and room volume calculation	Within approved design range
Pressure cascade	Room-to-room differential pressure under normal operation	Maintains directional cascade
Particle classification	Non-viable particle counts at defined locations	Meets applicable classification limit
Recovery	Time to return below target particle level after challenge	Within defined recovery time

4. Example results summary

Area	Test	Result	Assessment
Grade C room	HEPA integrity	Pass	All terminal filters acceptable
Grade C room	Differential pressure	12-15 Pa	Within 10-20 Pa target
Grade C room	Particle classification	Pass	At-rest classification acceptable
Grade C room	Recovery	11 min	Target <= 15 min
Grade D support	Differential pressure	7-10 Pa	Directional cascade maintained

5. Risk-based interpretation

The report should not treat individual test results in isolation. The qualification decision should confirm that airflow control, filtration, pressure direction and cleanroom classification collectively support the intended contamination-control strategy for the defined room use.

6. Qualification conclusion

The fictional results support qualification of the defined rooms and HVAC system for the intended operating state. Continued suitability depends on approved environmental monitoring, filter maintenance, pressure alarm management, periodic classification and change control.

7. Typical supporting records

- Room data sheets and airflow schematics
- HEPA integrity test sheets
- Airflow and differential-pressure raw data
- Particle-count printouts
- Recovery study data
- Calibration records
- Deviation assessments and final traceability review